

Name: Hamza Al-Sheikh

ID: 20464317

Abstract:

As mobile networks continue to evolve rapidly, 4G and 5G have emerged as two key terms in the realm of wireless communication. While both technologies aim to connect devices to the internet, they exhibit distinct differences in speed, latency, capacity, and coverage. Understanding these differences is crucial for consumers, businesses, and network providers, as the choice between 4G and 5G can influence everything from basic phone calls and messaging to the operation of smart cities and autonomous vehicles.

Introduction:

4G is a mobile network technology that improves upon its predecessor, 3G, by offering significantly faster data download speeds and enhanced internet access. It provides broadband cellular network services and adheres to standards set by the International Telecommunication Union (ITU-R), known as International Mobile Telecommunications Advanced (IMT-Advanced). 4G is designed to deliver services with bandwidth exceeding 100 Mbps, enabling the streaming of high-quality multimedia content.

Development of 4G began in 2000 and concluded in 2010, incorporating technologies such as LTE and WiMax. It employs Code Division Multiple Access (CDMA) with bandwidth capabilities ranging from 2000 Mbps to 1 Gbps. The primary goals of 4G include delivering high quality, speed, and capacity while enhancing security and reducing the costs associated with voice and data services, multimedia, and Internet Protocol (IP) communications. Additionally, it maintains compatibility with 2G and 3G networks.

5G represents the fifth generation of wireless network technology and is built on millimeter wave technology, which operates within a high-frequency spectrum ranging from 20 GHz to 96 GHz. Currently under development, this latest iteration of cellular technology promises significantly faster speeds than 4G, along with increased reliability and reduced response times. 5G utilizes Orthogonal Frequency-Division Multiplexing (OFDM) and millimeter wave technology, achieving data rates of up to 20 Gbps within frequency bands of 2-8 GHz. It is designed as a packet-based network capable of reaching speeds of up to 1000 Mbps or even 2.1 Gbps. This advanced technology supports high-quality video streaming, precise location tracking, low-latency communication, and improved real-time analytics capabilities. Additionally, it offers enhanced management features.

Importance:

We aim to provide a comparative study of 5G and 4G technologies using the OMNeT++ simulator, analyzing key performance metrics such as latency and throughput. This study helps in understanding how 5G improves upon the limitations of 4G and highlights the new features introduced in 5G. Different applications have varying requirements (e.g., high bandwidth for video streaming versus low latency for gaming). A comparative study assists in evaluating which technology is better suited for specific applications. This study is considered a valuable educational resource for students and professionals looking to understand the evolution of mobile communication technologies.

Research Question:

In this study, we seek to answer the question: What is the difference between 4G and 5G technologies?

Structure of the Report:

The report contains the following elements:

- **Introduction:** In this section, we present an overview of today's mobile networks, along with the research objective and research question.
- **Literature Review:** In this section, we review a collection of previous studies that we have analyzed and examined.
- **Network Design:** In this section, we outline our network design for both the fourth and fifth generations.
- **Results:** In this section, we present the results we have reached in the research.
- **Discussion:** In this section, we discuss the results and draw conclusions from the research findings.

Literature review:

In the literature review section, we aim to provide a comprehensive overview of existing research related to our research.

Background:

Mobile cellular data networks have enabled users to stay connected to the Internet while on the go. Various companies integrate this technology into their products, with notable examples being smart meters in homes and Tesla vehicles that receive updates over the air. Both of these applications utilize 4G and 5G technologies. This report will provide a technical overview of the technologies and protocols, particularly LTE Advanced, used in 4G and 5G networks, detailing how these technologies deliver services to users and facilitate data

transfer within the networks. Additionally, we will explore the distinct aspects of network architecture between 4G and 5G systems, as well as some of the challenges each faces.

Fourth Generation:

The fourth-generation mobile communication system, or 4G, encompasses TD-LTE and FDD-LTE, which together represent an evolution from the third-generation mobile communication system (3G) and WLAN technologies. It is important to note that LTE itself is essentially an upgrade of 3G, and it hasn't fully met the 4G criteria in its initial iterations. LTE Advanced, however, is the version that complies with the International Telecommunication Union's definition of 4G.

4G offers rapid data transmission along with high-quality video, audio, and images. It achieves impressive speeds, with a standard defining 4G as capable of data transfer rates exceeding 100 Mbps, translating to download speeds of approximately 12.5 MB/s to 18.75 MB/s. This performance is roughly twenty times faster than traditional home broadband, making it suitable for the vast majority of users' wireless service needs. 4G has transformed communication, entertainment, and business for many, providing unparalleled advantages over previous generations of technology. Unlike earlier communication technologies, 4G can be deployed in areas lacking DSL and CATV modem coverage, extending connectivity to a wider geographic region. In essence, 4G and LTE technology are synonymous, both having evolved from 3G wireless communication systems; thus, 4G can be viewed as an advanced version of 3G.

The key protocol underpinning 4G communication technology, LTE Advanced, consists of two main standards: TD-LTE and FDD-LTE. These correspond to different duplexing modes: Time Division Duplex (TDD) and Frequency Division Duplex (FDD). TDD allows for uplink and downlink transmissions to alternate in the same frequency band based on time allocation, optimizing spectrum utilization and layout. Conversely, FDD enables simultaneous uplink and downlink transmissions on separate frequency bands, enhancing data transmission capabilities. In essence, 4G employs OFDM for downlink modulation and SC-OFDM for uplink modulation, differentiating it significantly from 3G.

Fifth Generation:

The formal designation of 5G is IMT-2020, representing the fifth generation of mobile communication standards established by the International Telecommunication Union (ITU). Unlike previous generations which featured multiple competing standards, the development of the 5G protocol has established a cohesive industry-wide standard. The creation of 5G is the culmination of collaborative efforts from various organizations, with the 3rd Generation Partnership Project (3GPP) being the key player in standardization. Originally formed to produce technical reports and specifications for third-generation mobile communication systems, 3GPP has evolved into the leading authority for standard development. On average, the organization takes 1 to 2 years to define the specifications for a new version, with the initial version of 5G known as Release 15 (R15).

Release 15, as the inaugural 5G standard from 3GPP, is primarily focused on the Phase 1 aspects of 5G. This phase involves defining the New Radio (NR) framework—a new global standard for 5G based on Orthogonal Frequency Division Multiplexing (OFDM), which serves

as the foundation for next-generation cellular networks. Key elements addressed in this framework include the NR waveform, channel coding flexibility, and frame structure. Solutions for uplink and downlink decoupling, as well as the separation of Control Units (CU) and Distributed Units (DU), are specified within the architecture, along with clearly defined service types for ultra-Reliable Low Latency Communications (uRLLC).

In Phase 2 of the protocol, represented by R16, new multiple access technologies for NR will be explored, leading to further enhancements of the NR protocol framework established in R15. Beyond optimizing uRLLC based on R15 standards, R16 aims to differentiate itself with competitive advantages across various vertical industries. Below is a chart that outlines these developments.

Network traffic analysis under emerging beyond-5G scenarios for multi-band optical technology adoption

(Ruiz et al., 2023)

Problem Statement:

The ITU-T Focus Group Network 2030 is guiding network operators in identifying the requirements and use cases that networks must fulfill over the current decade. However, network operators face the challenge of evolving their networks to meet stringent performance requirements, including the need for large bandwidth to support anticipated beyond-5G (B5G) services such as digital twins and volumetric video. To achieve this bandwidth sustainably and scalably, there is a need to transition from legacy optical networks to multi-band (MB) optical networks that can utilize frequency bands beyond the traditional C + L bands.

Solution Approach:

To address this problem, the authors propose a traffic analysis methodology that enables network operators to compute the expected traffic demand resulting from the integration of established mass market services with anticipated B5G service scenarios. By leveraging numerical results derived from inputs and forecasts provided by major European network operators, they demonstrate that multi-band optical networks will be essential across all segments of the network—including metro-aggregation, metro-core, and backbone—by the end of this decade. This approach will assist network operators in planning and optimizing their networks to meet future demands effectively.

Performance Analysis of a 5G Transceiver Implementation for Remote Areas Scenarios

(Dias et al., 2018)

Problem Statement:

The fifth generation (5G) mobile communication networks are expected to support a wide range of new services and applications, one of which is providing broadband Internet access to remote areas. This is particularly critical as a significant portion of the global population residing in sparsely populated regions lacks Internet connectivity. Improving communication infrastructure in these rural areas can have substantial social and economic benefits, particularly in enhancing agricultural productivity. However, there is a need to analyze the performance of specialized transceiver technologies designed for these remote environments to ensure they meet the required standards for effective communication.

Solution Approach:

This paper presents an analysis of a Software-Defined (SD) transceiver specifically designed for remote area coverage, implemented on field-programmable gate array (FPGA) hardware for real-time processing capabilities. The transceiver utilizes advanced digital communication techniques, including Generalized Frequency Division Multiplexing (GFDM) combined with a 2x2 Multiple-Input Multiple-Output (MIMO) diversity scheme and polar channel coding.

To evaluate the performance of the prototype, we assess its out-of-band emissions and bit error rate (BER) under an Additive White Gaussian Noise (AWGN) channel. The results will provide insights into the effectiveness of this transceiver design for delivering reliable broadband Internet access in remote areas, ultimately supporting the goal of improving connectivity and productivity in rural communities.

The use of 5G Non-Public Networks to support Industry 4.0 scenarios.

(Ordóñez-Lucena et al., 2019)

Problem Statement:

The ongoing digital transformation is essential for advancing towards Industry 4.0, characterized by more efficient, sustainable, and connected industrial systems. However, the implementation of Industry 4.0 presents unique challenges, particularly in logistics and manufacturing sectors. These challenges include stringent requirements for low latency, high reliability, and precise positioning in industrial processes. Traditional use cases from other sectors do not face such demanding criteria, making it crucial to explore how 5G technology can meet these needs in indoor, private environments.

The emergence of 5G non-public networks offers a potential solution to address these challenges. However, there is a lack of comprehensive understanding regarding their applicability within the Industry 4.0 ecosystem, as well as the various deployment options available and their integration with public mobile networks.

Solution Approach:

This article provided an overview of 5G non-public networks and their relevance to the Industry 4.0 landscape. It analyzes the ongoing developments in the 3GPP Rel-16 specifications to identify several deployment options for non-public networks. The authors discussed how these options can be integrated with existing public networks operated by mobile network providers. Furthermore, the authors conduct a comparative analysis of the identified deployment options based on several criteria, including technical feasibility, regulatory considerations, and business implications. This analysis aims to equip industry

stakeholders with the necessary insights to determine the most suitable deployment option for their specific use cases within the Industry 4.0 framework.

The findings of this study will assist companies in logistics and manufacturing sectors in making informed decisions regarding the adoption of non-public networks, ultimately enhancing their operational flexibility, productivity, and usability in industrial processes.

Network Design:

In this section, we provide the architecture of 4G technology and 5G technology. Then we describe the network that we design at OMNET++ simulator.

5G Architecture:

The gNB node is responsible for terminating the NR user plane and control plane protocols for user equipment (UE), which includes devices such as smartphones, tablets, and laptops. It connects to the 5G Core (5GC) via the NG interface. Similarly, the ng-eNB node handles the E-UTRA (LTE) user plane and control plane protocol terminations for UE and also connects to the 5GC through the NG interface. The Access and Mobility Management Function (AMF) and User Plane Function (UPF) are key components in this architecture.

The 3GPP TS 38.401 standard outlines the user and control planes of 5G NR as well as the RAN architecture, detailing interfaces such as NG, Xn, and F1, along with their interactions with the radio interface (Uu).

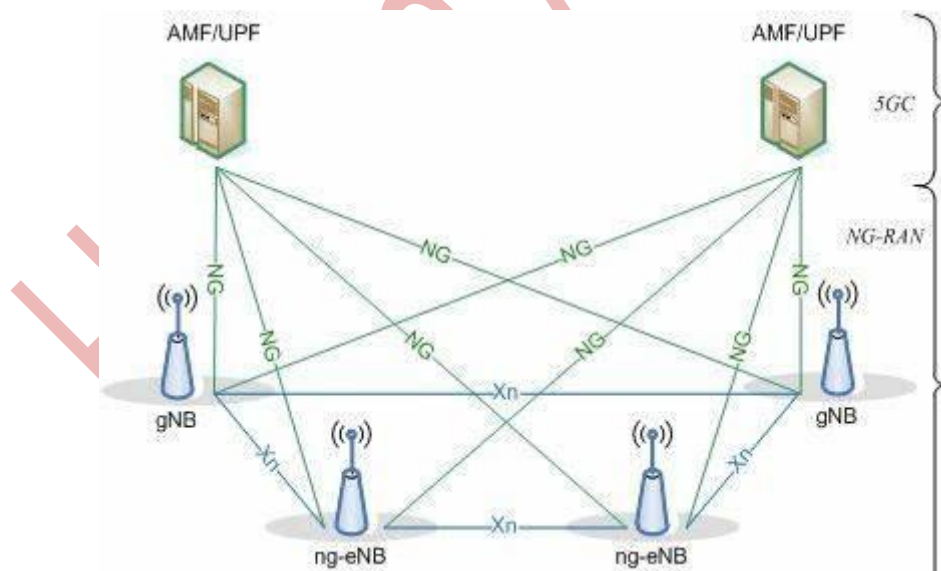


Figure (1) 5G Architecture

5G smartphones communicate with the 5G RAN using the Uu radio interface, while the 5G RAN connects to the 5G Core Network (5GC).

4G Architecture:

LTE, or Long Term Evolution, originated as a project in 2004 under the guidance of the telecommunications organization known as the Third Generation Partnership Project (3GPP). The corresponding evolution of the GPRS/3G packet core network is referred to as System Architecture Evolution (SAE). The term LTE is commonly used to encompass both LTE and SAE.

LTE is an advancement of an earlier 3GPP system called the Universal Mobile Telecommunications System (UMTS), which itself evolved from the Global System for Mobile Communications (GSM). Related specifications are officially designated as evolved UMTS terrestrial radio access (E-UTRA) and evolved UMTS terrestrial radio access network (E-UTRAN). The first version of LTE was documented in Release 8 of the 3GPP specifications.

The primary objective of LTE is to deliver high data rates, low latency, and packet-optimized radio access technology that supports flexible bandwidth deployment. Additionally, its network architecture is designed to accommodate packet-switched traffic while ensuring seamless mobility and high quality of service.

The architecture of a 4G network system is divided into three primary components: the EPC, eNB, and UE. Collectively, the EPC and the radio access network E-UTRAN form what is commonly referred to as the 4G network. The core network, EPC, plays a crucial role in verifying user identities, transmitting user data, and managing that data. Meanwhile, the radio access portion, E-UTRAN, encompasses various functions performed by eNB and NodeB, as well as certain capabilities of the core network itself.

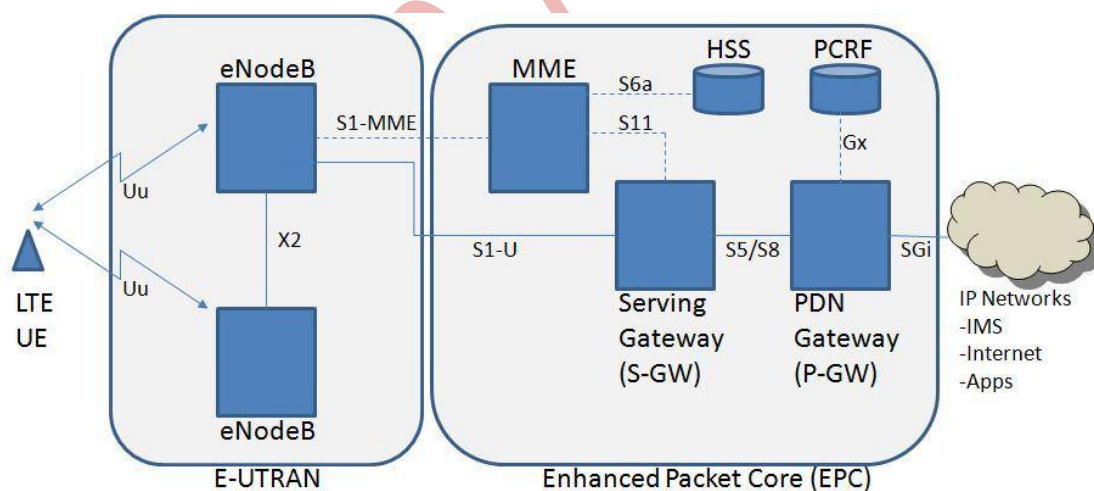


Figure (2) 4G Architecture

The high-level architecture of LTE consists of three primary components:

- User Equipment (UE): The internal structure of LTE user equipment mirrors that of UMTS and GSM, which is classified as Mobile Equipment (ME).
- Evolved UMTS Terrestrial Radio Access Network (E-UTRAN): The architecture of the Evolved UMTS Terrestrial Radio Access Network (E-UTRAN) is designed to manage

radio communications between mobile devices and the Evolved Packet Core. It comprises a single component: the evolved base stations, referred to as eNodeB or eNB. Each eNB is responsible for controlling mobile devices within one or more cells, with the base station communicating with a mobile device known as its serving eNB.

- Evolved Packet Core (EPC): The architecture of the Evolved Packet Core (EPC) is depicted below. To maintain clarity, several additional components have been omitted from the diagram. These include systems such as the Earthquake and Tsunami Warning System (ETWS), the Equipment Identity Register (EIR), and the Policy Control and Charging Rules Function (PCRF).

The Evolved Packet Core interfaces with external packet data networks, such as the internet, private corporate networks, or the IP multimedia subsystem.

In LTE, a mobile device connects to only one base station and one cell at any given time. The eNB supports two main functions:

- The eNB transmits and receives radio signals to and from all mobile devices using both analog and digital signal processing functions of the LTE air interface.
- The eNB manages the low-level operations of all connected mobiles by sending signaling messages, such as handover commands.

Each eNB connects to the EPC via the S1 interface and can also link to neighboring base stations through the X2 interface, primarily used for signaling and packet forwarding during handovers.

A home eNB (HeNB) is a base station purchased by a user to provide femtocell coverage at home. It is part of a closed subscriber group (CSG) and can only be accessed by mobiles with a USIM that belongs to that group.

OMNET++ Simulator:

OMNET++ Network Simulator is an object-oriented, modular discrete event network simulation framework with a generic architecture, making it suitable for a wide range of applications. Key features of OMNET++ include its simulation kernel, graphical network editors for NED files, tools for plotting data, and a compiler for the NED topology description languages. Additionally, it offers various utilities such as a random number seed generation tool and a make file creation tool. Users can execute simulations through two types of interfaces: a command-line user interface and a graphical user interface.

Performance Metrics:

Performance metrics are quantitative measures used to evaluate the effectiveness, efficiency, and success of a particular network.

Latency: refers to the time delay from the request is made until the response is received. It is often measured in milliseconds (ms)

Throughput: measures the amount of data processed or transmitted in a given amount of time.

Scenarios:

We design two scenarios, one with 4G network (LTE), and the second with 5G network (NR). At both we use the same parameters. 100 UE devices, with 1000x600 m, and video conference application.

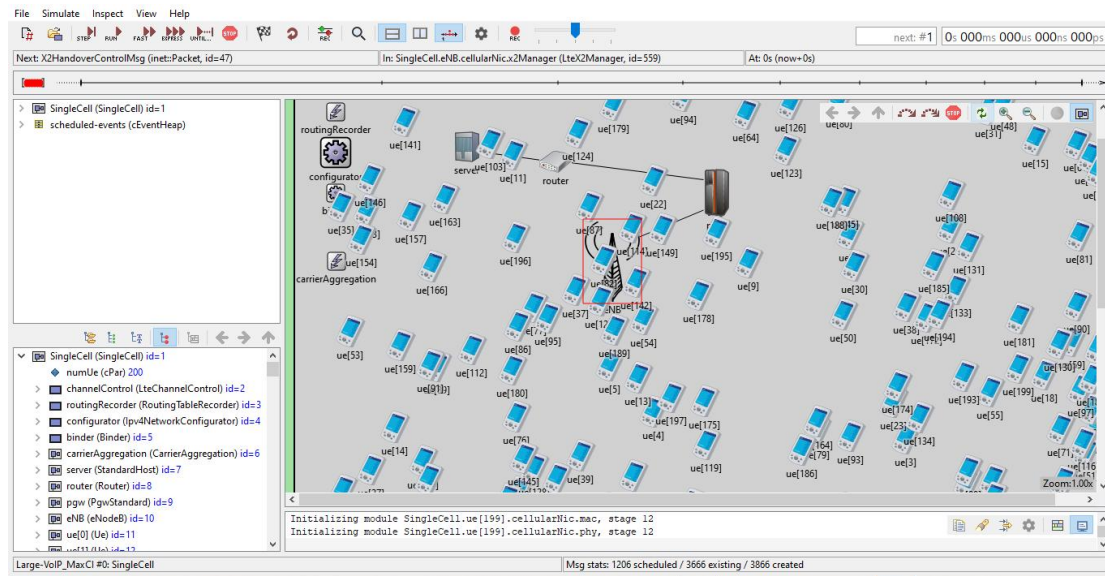


Figure (3) network

Result:

In this section, we illustrate the values and results. we compare two key performance metric latency and throughput of 4G and 5G mobile networks.

	4G	5G
Latency	0.9 msec	0.006 msec
Throughput	694.8 Mbps	1.3 Gbps

Result Table

The latency for 4G networks is 0.9 milliseconds (ms). This indicates the time it takes for data to travel from the user device to the network and back. the latency for 5G networks is dramatically lower at 0.006 milliseconds (ms). This significant reduction means that 5G can support applications requiring near-instantaneous data transfer

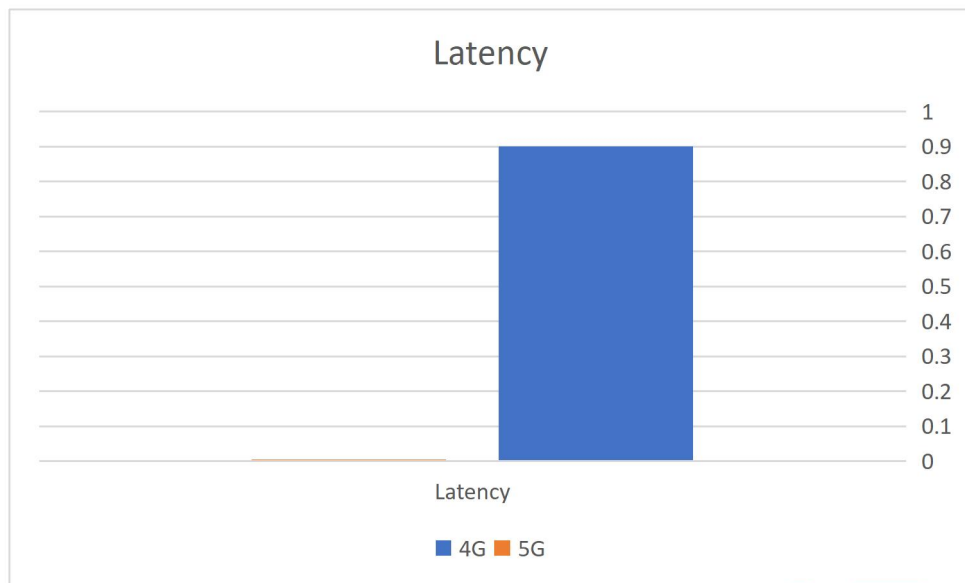


Figure (4) Latency

The throughput for 4G networks is 694.8 megabits per second (Mbps). This represents the maximum rate at which data can be transmitted over the network, affecting activities like streaming video or downloading files. The throughput for 5G networks is substantially higher at 1.3 gigabits per second (Gbps), which is equivalent to 1300 Mbps. This increase allows for faster downloads, smoother streaming of high-definition content, and better performance for applications that require large amounts of data.

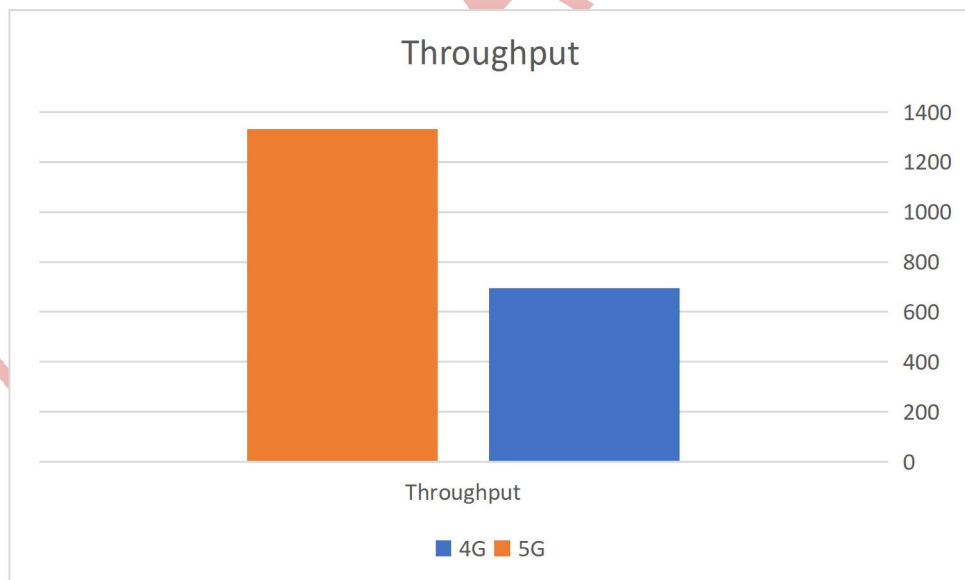


Figure (5) throughput

Discussion:

we compare the performance metrics of 4G and 5G networks with 100 User Equipment (UE) devices in each scenario. in this section we provide analysis and discussion of these metrics: latency and throughput.

Latency

The latency in the 5G scenario is significantly lower than in the 4G scenario, demonstrating a substantial improvement. A reduction from 0.9 ms to 0.006 ms represents a nearly 150-fold decrease in latency.

Throughput

The throughput of 5G is also significantly higher than that of 4G, nearly doubling the capacity from 694.8 Mbps to 1.3 Gbps.

This increased throughput allows for more simultaneous connections and higher data rates per user, which is crucial in densely populated areas where many devices are connected at once.

With higher throughput applications that require large amounts of data transfer, such as high-definition video streaming and cloud gaming, can be supported more effectively without buffering or interruptions.

Conclusion:

The results clearly illustrate that 5G technology offers significant advantages over 4G in terms of latency, and throughput when supporting a similar number of devices. This transition not only enhances user experiences but also paves the way for innovative applications and services that require high-performance networks. As the world becomes increasingly connected, the role of efficient and high-capacity networks like 5G will be pivotal in shaping future communication landscapes.

While 4G continues to adequately serve many users and applications, 5G represents a significant leap forward in mobile technology, offering the capabilities necessary to support the next generation of communication, automation, and technological innovation. As 5G adoption grows, it is expected to unlock new possibilities across various sectors, transforming how we connect and interact with the digital world.

References:

Agbo, P. and Weitkemper, P. (2023). Analysis of Different MEC Offloading Scenarios with LEO Satellite in 5G Networks. *2023 IEEE International Conference on Omni-layer Intelligent Systems (COINS)*, pp.1–6. doi:<https://doi.org/10.1109/coins57856.2023.10189256>.

Ruiz, M., Hernández, A., Quagliotti, M., Emilio Hugues Salas, Riccardi, E., Rafel, A., Velasco, L. and Gonzalez, O. (2023). Network traffic analysis under emerging beyond-5G scenarios for multi-band optical technology adoption. *Journal of Optical Communications and Networking*, 15(11), pp.F36–F36. doi:<https://doi.org/10.1364/jocn.492128>.

Dias, W., Gaspar, D., Mendes, L., Chafii, M., Matthe, M., Neuhaus, P. and Fettweis, G. (2018). Performance Analysis of a 5G Transceiver Implementation for Remote Areas Scenarios. *2018 European Conference on Networks and Communications (EuCNC)*, pp.363–367. doi:<https://doi.org/10.1109/eucnc.2018.8443268>.

Ordóñez-Lucena, J., Chavarria, J.F., Contreras, L.M. and Pastor, A. (2019). *The use of 5G Non-Public Networks to support Industry 4.0 scenarios*. [online] IEEE Xplore. doi:<https://doi.org/10.1109/CSCN.2019.8931325>.

Abstract:	1
Introduction:	1
2	Importance:
2	Research Question:
2	Structure of the Report:
Literature review:	2
2	Background:
3	Fourth Generation:
Network Design:	6
6	5G Architecture:
7	4G Architecture:
8	OMNET++ Simulator:
8	Performance Metrics:
9	Scenarios:
Result:	9
Discussion:	11
11	Latency
11	Throughput
Conclusion:	11
References:	12